

**A Mini Project Submitted In Partial Fulfilment
Of The Requirements For the degree of**

Bachelor of Technology

In

**Computer Science & Engineering
(Artificial Intelligence & Machine Learning)**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine Learning)**

**NOIDA INSTITUTE OF ENGG. & TECHNOLOGY, GREATER NOIDA, GAUTAM BUDDH NAGAR
(AN AUTONOMOUS INSTITUTE)**

(Approved by AICTE and affiliated to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow)



2024-2025

Certificate

I hereby certify that the work which is being submitted in the Project Report entitled “**Nidhi Kumari**” in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Science and Engineering Artificial Intelligence & Machine Learning** and submitted to the Department of Computer Science & Engineering **Artificial Intelligence & Machine Learning, Noida Institute of Engineering & Technology, Greater Noida** is an authentic record of my workshop carried out during Fourth semester under the supervision of **Ms. Meghali Das (Assistant Professor)**, Department of Computer Science and Engineering **Artificial Intelligence, Noida Institute of Engineering & Technology, Greater Noida**. The matter embodied in this project Report is original and has not been submitted for any other degree or diploma award.

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DECLARATION

I hereby declare that this submission is my project and that, to the best of my knowledge and belief, it contains no material previously published by another person nor material which, to a substantial extent, has been accepted for the award of any other degree or diploma of the university or other institute of higher learning except where due acknowledgment has been made in the text.

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Acknowledgement

I would like to express my sincere gratitude to all those who provided their valuable support and guidance throughout the successful completion of this mini project titled:

“JOBORA – An Online Job Portal for Connecting Job Seekers with Employers”

I extend my heartfelt thanks to **Ms. Meghali Das (Assistant Professor)**, Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning), **Noida Institute of Engineering & Technology, Greater Noida**, for her constant support, valuable suggestions, and expert supervision throughout the duration of this project. Her encouragement and constructive feedback helped me in every stage of the work.

I am also thankful to **Dr. Raju**, Head of the Department, for providing all the necessary resources and infrastructure that facilitated the smooth execution of the project.

Lastly, I would like to thank my peers and everyone who directly or indirectly contributed to the successful completion of this project.

This project has helped me enhance my understanding of Database Management Systems and has given me valuable practical exposure to real-world applications.

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1.Abstract

This project aims to develop an online job portal named **JOBORA** that efficiently connects job seekers with potential employers. The system provides a platform where job seekers can create profiles, upload resumes, search for jobs based on skill, location, and experience, and apply for suitable positions. On the other side, employers can register, post job openings, review applications, and manage hiring pipelines. This portal is entirely database-driven and uses a **relational DBMS (MySQL)** to store and manage large volumes of structured data securely and efficiently. The design ensures data consistency, eliminates redundancy through normalization, and establishes meaningful relationships between entities such as users, jobs, and applications. The system uses **Entity-Relationship (ER) modeling** during the planning phase to design a well-structured schema, and **SQL queries** are used to create, update, retrieve, and delete data as needed. Advanced DBMS features such as indexing, joins, foreign keys, and constraints are also integrated to enhance performance and maintain data integrity. Moreover, the project emphasizes the importance of using **DBMS principles** in real-life applications by simulating scenarios like real-time job searches, multi-user environments, and user access management. The end result is a **fully functional prototype** that can be scaled or integrated into actual recruitment workflows. JOBORA serves as a practical example of how database technologies can be applied to solve real-world challenges in the employment and HR sector.

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2. Introduction

With the rise of digital platforms, job recruitment has transformed into a fast and data-driven process. Traditional methods like newspaper ads, walk-ins, and offline job boards have mostly been replaced by online job portals that offer wider reach, quicker updates, and smart filtering of candidates.

JOBORA is a simplified version of such a digital job portal, designed mainly to show how backend and database systems work. Instead of focusing on the front-end design or external tools, JOBORA highlights how data is stored, organized, and managed using a **relational database management system (RDBMS)**.

The project demonstrates important **DBMS concepts** such as:

- **Entity-Relationship (ER) modeling** to plan a strong database structure.
- **Normalization** to reduce data repetition and improve efficiency.
- **Foreign keys and joins** to connect users, job posts, and applications.
- **SQL queries** to add, update, and fetch data from the database.

Key functions of JOBORA include:

- **User registration and login** with different roles (job seeker or employer).
- **Job posting** by employers and **application submission** by job seekers.
- **Search filters** based on skills, location, and experience.
- **Dashboards** to help employers review applicants and job seekers track their application status.

By simulating real-world hiring features like resume management, job matching, and application tracking, **JOBORA** shows how a well-planned DBMS can support powerful, real-time recruitment platforms. It serves as both a learning and practical tool for understanding backend development in employment services.

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3. Technology Background

The JOBORA project is developed using fundamental principles of database design and management, aimed at creating a secure, efficient, and scalable backend system for job seekers and employers. This section explores the technological background of job portal systems, highlighting traditional approaches and modern digital platforms.

1. Traditional Job Recruitment Systems

Traditional job recruitment relied on offline processes and manual record-keeping. These methods, although foundational, had various limitations:

- Depended on newspaper advertisements and physical applications.
- Manual shortlisting and scheduling led to delays.
- Prone to human errors and data mismanagement.
- Difficult to maintain candidate and job posting records.

2. Existing Digital Systems

To overcome these challenges, digital job portals emerged, especially for medium to large-scale hiring needs. These systems introduced automation but often lacked customization and backend depth for academic or small business purposes:

- Basic portals with job listings and resume uploads.
- Limited features for employer control and candidate tracking.
- Some allow user logins and email notifications.
- Lack advanced filtering, analytics, and backend optimization.

4. Objective

The goal of the **JOBORA** project is to create a reliable and scalable database system for a job portal that connects employers and job seekers. This system will use important **DBMS concepts** to organize, manage, and protect the data involved in the recruitment process.

Project Goals:

1. Design a Simple and Efficient Database Structure

- Build a database that reflects how the job portal works.
- Use normalization (up to 3NF) to reduce duplicate data and make sure the data is consistent.
- Create an **ER model** that shows key entities (like Job_Seeker, Employer, Job_Post, and Application) and their relationships.

2. Create Tables and Set Relationships Using SQL

- Set up tables with **primary and foreign keys** to maintain data integrity.
- Define **data types and constraints** (like NOT NULL, UNIQUE, DEFAULT) to ensure accurate data.
- Set up **one-to-many and many-to-many relationships** (e.g., one employer can post many jobs, and one job can have many applicants).

3. Let Users Interact with the Database (Conceptually)

- Imagine how users (job seekers and employers) would interact with the system through forms (for registration, job posting, applying).
- These actions will be linked to **SQL queries** (INSERT, UPDATE, DELETE) in the background.
- Show how user actions are translated into safe and effective database operations.

4. Include Key Functionalities:

- **Job Posting** – Employers can add new job listings with the necessary skills, qualifications, and locations.

- **Job Searching** – Job seekers can search for jobs by filtering based on keywords, location, or skills.
- **Job Application** – Job seekers can apply for jobs, and the system will track the application's status and time.
- **Profile Management** – Users can update their profiles or job posts with **SQL UPDATE** commands.

5. Use **Advanced SQL Queries for Data Management**

- Use **JOINS and subqueries** to combine data, such as showing all applications for a job with the details of the applicants.
- Perform **aggregation** (e.g., count how many applications each job has).
- Use **SQL commands** like ORDER BY, WHERE, and GROUP BY to sort, filter, and group the data.

6. **Simulate Real-World Job Portal Features**

- Show how job portals like LinkedIn or Indeed use databases.
- Create features like employer dashboards, candidate tracking, and job recommendations using **DBMS logic**.
- Show how the system can handle **multiple users** at the same time while keeping the data safe.

5. Importance Of Project

This project is significant for several reasons:

- **Improves Accessibility:** Job seekers can access and apply for jobs from anywhere, overcoming geographical barriers.
- **Saves Time:** Employers can post jobs and receive applications in real-time, which reduces the time spent on traditional hiring processes.
- **Promotes Efficiency:** The advanced search filters and personalized recommendations help job seekers quickly find relevant jobs, while employers can efficiently browse suitable candidates.
- **Reduces Human Error:** By automating the data entry and application processes, JOBORA minimizes the chances of mistakes such as data duplication or incomplete applications.
- **Scalability:** The database structure is designed to handle large volumes of users and job postings, enabling future scalability. Additional features like AI-based job recommendations, skill assessment tests, or analytics dashboards can be integrated as the platform grows.
- **Educational Value:** The project provides valuable hands-on experience in using key DBMS tools and concepts, such as relational databases, SQL queries, normalization, and data modeling. It also gives insight into building a real-world system that integrates various DBMS components for efficient and effective management.
- **Supports Data-Driven Decision Making:** Employers can generate reports based on applications, job views, and hiring patterns, which helps make informed decisions about recruitment strategies and budgeting.
- **Better Candidate Matching:** With a structured database, the system can be enhanced in the future to integrate machine learning algorithms for better matching of candidates to job roles based on their experience, skills, and preferences.

6. Technologies Used In Projects

TOOLS & TECHNOLOGIES USED

To build the foundation of **JOBORA**, we used reliable tools and techniques that ensured smooth data operations, logical structure, and real-world applicability. Below are the key components that powered the backend:

MySQL

Used as the core database management system. It stores structured data such as user accounts, job listings, and applications in relational tables. MySQL's support for features like indexing, constraints, and stored procedures made JOBORA efficient and scalable.

ER Diagrams

Before writing any SQL code, we used ER diagrams to design the data model visually. This helped us identify important entities (Job Seeker, Employer, Job, Application) and their relationships, ensuring clarity and structure from the start.

Normalization

To avoid data redundancy and improve performance, we applied normalization rules up to the third normal form (3NF). This helped create a clean, optimized database structure.

SQL (Structured Query Language)

SQL was used to create and manage the entire database. Different categories of SQL commands helped in various stages:

- **DDL** for defining tables and schemas.
- **DML** for managing records (insert, update, delete).
- **DCL** for handling user permissions.
- **TCL** for managing transactions during critical operations.

MySQL Workbench

A graphical tool that helped us model, design, and interact with the MySQL database. It was useful for running queries, visualizing schemas, and refining the database design during development.

7. Outcomes

	user_id	name	email	user_type
▶	1	Ravi Kumar	ravi.kumar@mail.com	seeker
	2	Priya Sharma	priya.sharma@tcs.com	employer
	3	Neha Verma	neha.verma@mail.com	seeker
	4	Anil Mehta	anil.mehta@infosys.com	employer
	5	Rajesh Iyer	rajesh.iyer@mail.com	seeker
*	NULL	NULL	NULL	NULL

	job_id	employer_id	category_id	title	location	salary	required_skills	job_type
▶	1	1	1	Data Analyst	Remote	50000	SQL, Excel, Data Analysis	Remote
	2	2	1	Frontend Developer	Chennai	40000	HTML, CSS, JavaScript	Full-time
	3	1	1	ML Developer	Hyderabad	70000	Python, Machine Learning, SQL	Full-time
	4	2	1	Java Developer	Bangalore	700000	Java, Spring	Full-Time
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	application_id	job_title	seeker_name	application_date	application_status
▶	1	Data Analyst	Ravi Kumar	2025-04-21	Approved
	2	ML Developer	Ravi Kumar	2025-04-21	Rejected
	4	Frontend Developer	Ravi Kumar	2025-04-21	Pending
	3	Frontend Developer	Rajesh Iyer	2025-04-21	Pending

	application_id	job_id	seeker_id	application_date	status
▶	1	1	1	2025-04-21	Approved
*	NULL	NULL	NULL	NULL	NULL

	job_id	employer_id	category_id	title	location	salary	required_skills	job_type
▶	4	2	1	Java Developer	Bangalore	700000	Java, Spring	Full-Time

	Message
▶	Application status updated succ...

8. Conclusion

The **JOBORA** project helped us understand how database systems can solve real-world problems in a structured and meaningful way. It focused fully on the backend using database tools, without relying on web development or interfaces.

Here's what the project achieved:

- ❖ **Built a Functional Database System**

Designed a complete system that stores and manages job seekers, employers, job posts, and applications.

- ❖ **Learned by Doing**

Applied DBMS topics like ER modeling, normalization, joins, and transactions directly—making learning hands-on and effective.

- ❖ **Organized Data with Clarity**

Used normalization to avoid repeated data and designed relationships that keep everything linked properly.

- ❖ **Used Constraints to Protect Data**

Implemented primary and foreign keys to ensure all data stays accurate, related, and secure.

- ❖ **Created Real-World-Like Scenarios**

Added job search filters, application tracking, and registration processes—just like in actual portals, all using SQL.

- ❖ **Handled Critical Operations Safely**

Used transaction controls like COMMIT and ROLLBACK to make sure important actions (like applying to a job) happen safely.

- ❖ **Prepared for Larger Systems**

The database structure is flexible and can be expanded to handle more features or more data without major changes.

- ❖ **Strengthened DBMS Understanding**

Gained strong confidence in using DBMS tools like MySQL, phpMyAdmin, and SQL to build something from the ground up.